LUBRICATION AND BEARINGS



A Lubrication is the Reduction of Friction to a Minimum by Replacing Solid Friction with Fluid Friction

Friction can be defined as the resistance to movement between any two surfaces in contact with each other.

The study of how surfaces interact in relative motion is called Tribology

LUBRICATION



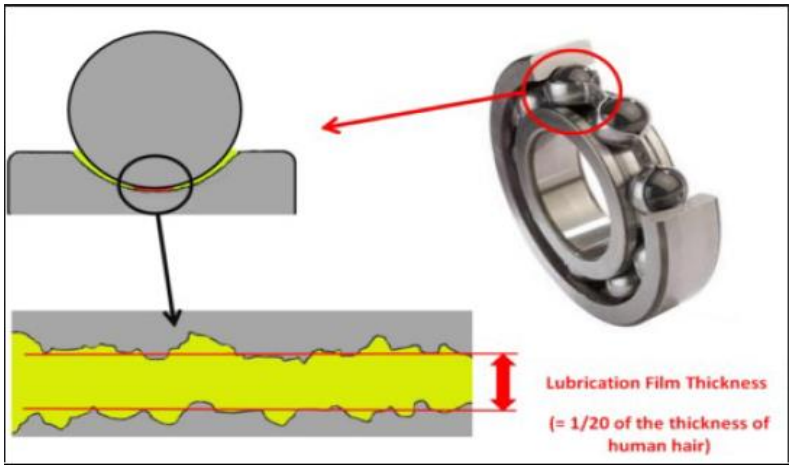
Lubrication *is the process, or technique employed to reduce wear of one or both surfaces in close proximity, and moving relative to each other by interposing a substance called* ***lubricant*** *between the surfaces to carry or to help carry the load (pressure generated) between the opposing surfaces.*

- The primary objective in lubrication is to reduce the severity of both the normal and shear stresses in solid surfaces contact that move relative to each other.

- One universal fact in the theories of friction and wear is that only a small fraction of the nominal area of contact between two bodies is in actual contact.

• The actual contact area may be as little as 0.01% of the apparent area of contact, and no stresses exist in the regions between.

• The stress-state in most of the actual areas of contact exceeds the yield point of ductile materials and the fracture strength of brittle materials.



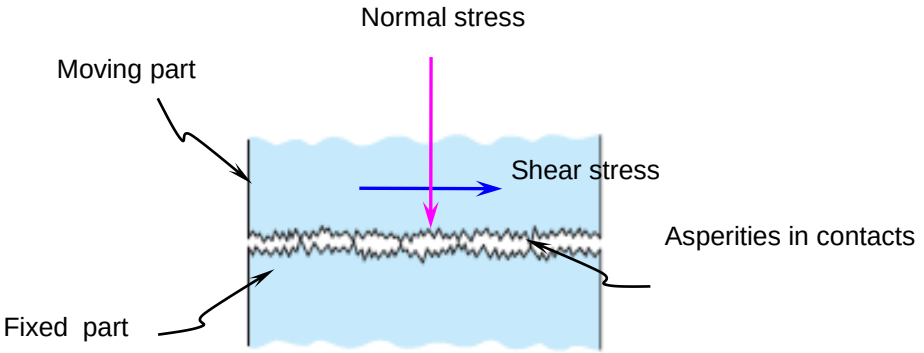


Fig.1a Solid surfaces

- All of the mechanical energy applied to (absorbed by) an un-lubricated bearing heats and deforms the sliding surfaces.

• The adverse effects of contact between rough surfaces can be reduced by smoothing out the variations in surface stress to some lower average values.

• Smoothing may be accomplished by inserting any substance between solid surfaces that minimizes friction and wear and the substance is a known as lubricant.

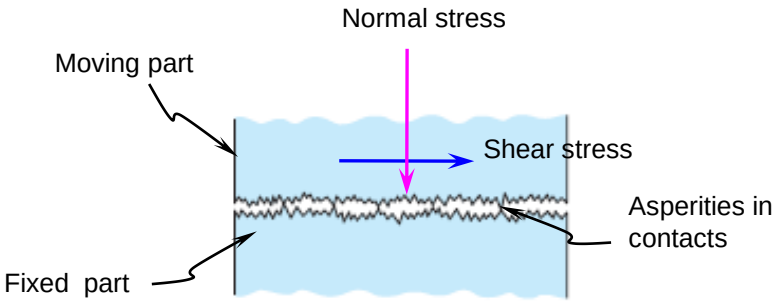


Fig.1a Solid surfaces

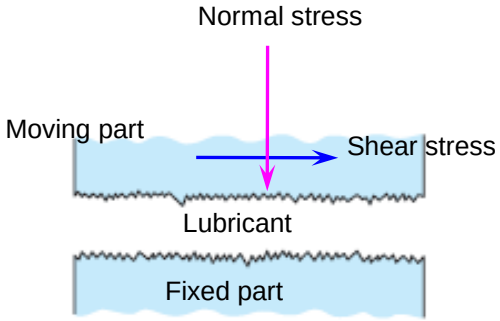


Fig.1b Introduction of lubricant

➢ When there are hard contaminants in the oil, and they are greater in size than average lube film, then 3 body abrasive wear starts to happen. This then leads to surface indentation and scratching, which is a process that can lead to bearing or component failure.

- Examples:
  - In a sleeve bearing, a shaft, or journal, rotates or oscillates within a sleeve, or bushing, and the relative motion is sliding.
  - In an antifriction bearing, the main relative motion is rolling.
  - A follower may either roll or slide on the cam.
  - Gear teeth mate with each other by a combination of rolling and sliding.
  - Pistons slide within their cylinders.

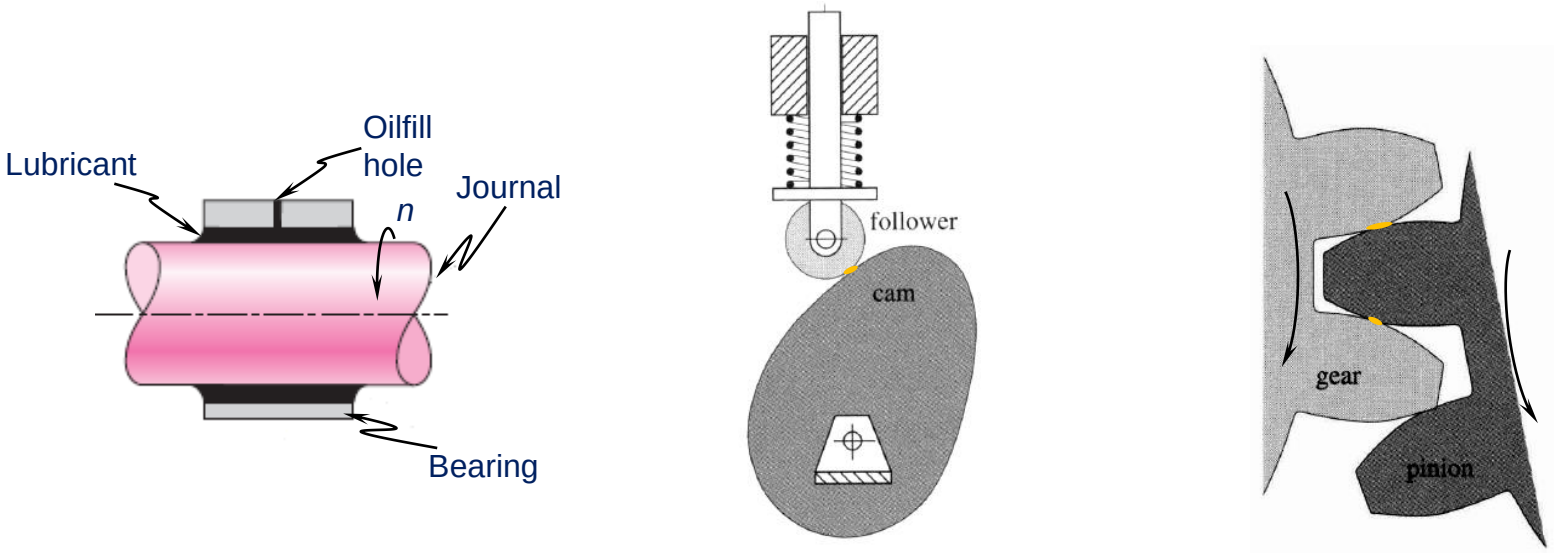


Fig. 1c Examples of bearings

- Five distinct forms of lubrication may be identified as:

1) Hydrodynamic lubrication
2) Hydrostatic lubrication
3) Elasto-hydrodynamic lubrication
4) Boundary lubrication
5) Solid film lubrication

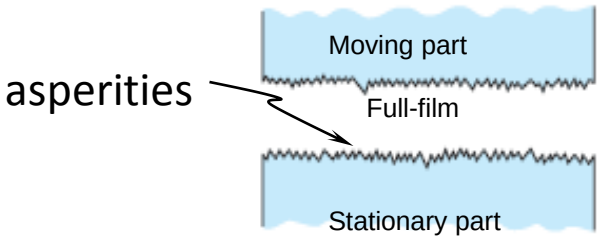


Fig.2 Hydrodynamic lubrication

1. Hydrodynamic lubrication

We know that a surface will have tiny asperities or peaks that will contact if two plates are placed together. If one of the plates were to slide over the other, then friction would increase, the asperities would break and the surfaces would wear. In hydrodynamic lubrication, *a fluid film separates the surfaces, prevents wear and reduces friction*.

Hydrodynamic lubrication

- Also called fluid-film, thick-film, or flooded lubrication
- A thick film of lubricant is interposed between the surfaces of bodies in relative motion
- The pressure buildup takes place in the film due to **relative motion of the surfaces**
- Fluid friction is substituted for sliding friction
- Coefficient of friction is decreased
- Prevalent in journal and thrust bearings

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Hydrodynamic lubrication

- The load-carrying surfaces of the bearing are separated by a *relatively thick film of lubricant*, so as *to prevent metal-to-metal contact*, and that the stability thus obtained can be explained by the *laws of fluid mechanics*.

- The film pressure is created by the moving surface itself pulling the lubricant into a wedge-shaped zone at a velocity sufficiently high to create the pressure necessary to separate the surfaces against the load on the bearing.

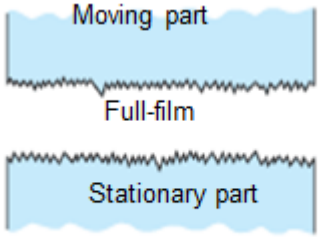


Fig.2 Hydrodynamic lubrication

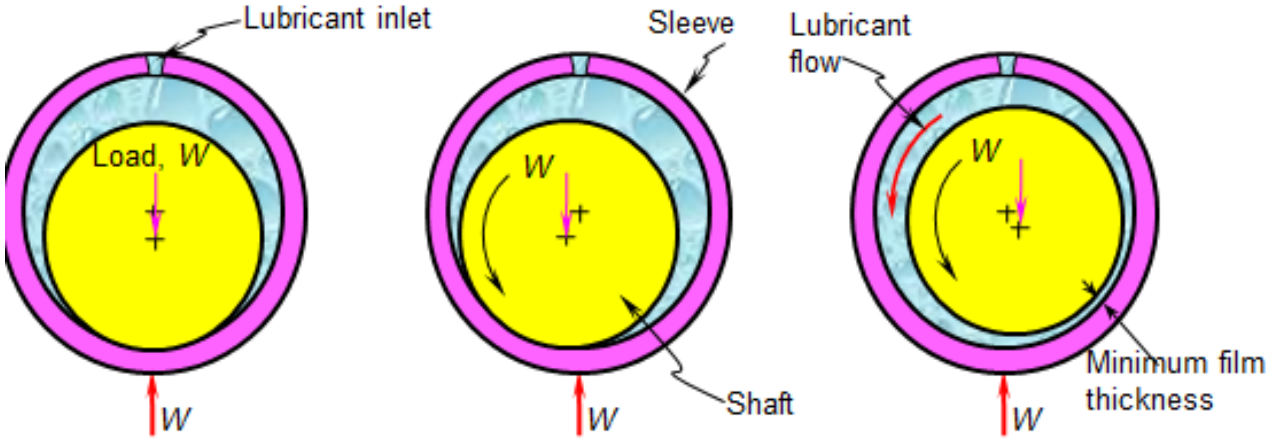
Application of hydrodynamic lubrication

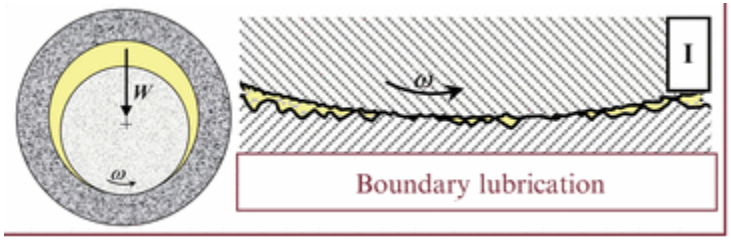
Large plain bearings like pedestal bearings, main bearing of diesel engines.

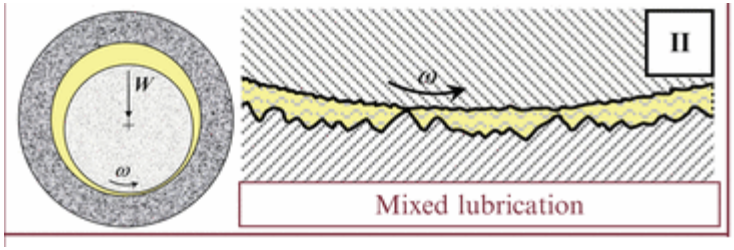
Delicate instruments. Light machines, like watches, clocks, guns, sewing machines. Scientific instruments.

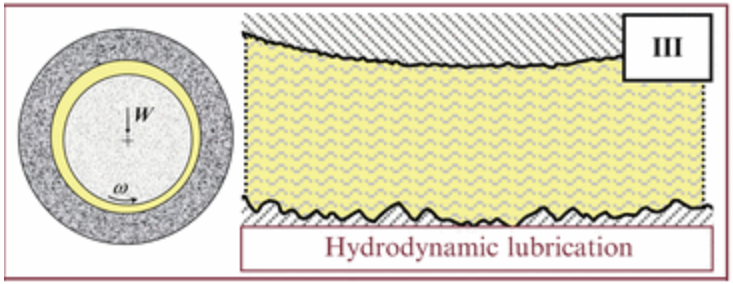
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Hydrodynamic lubrication

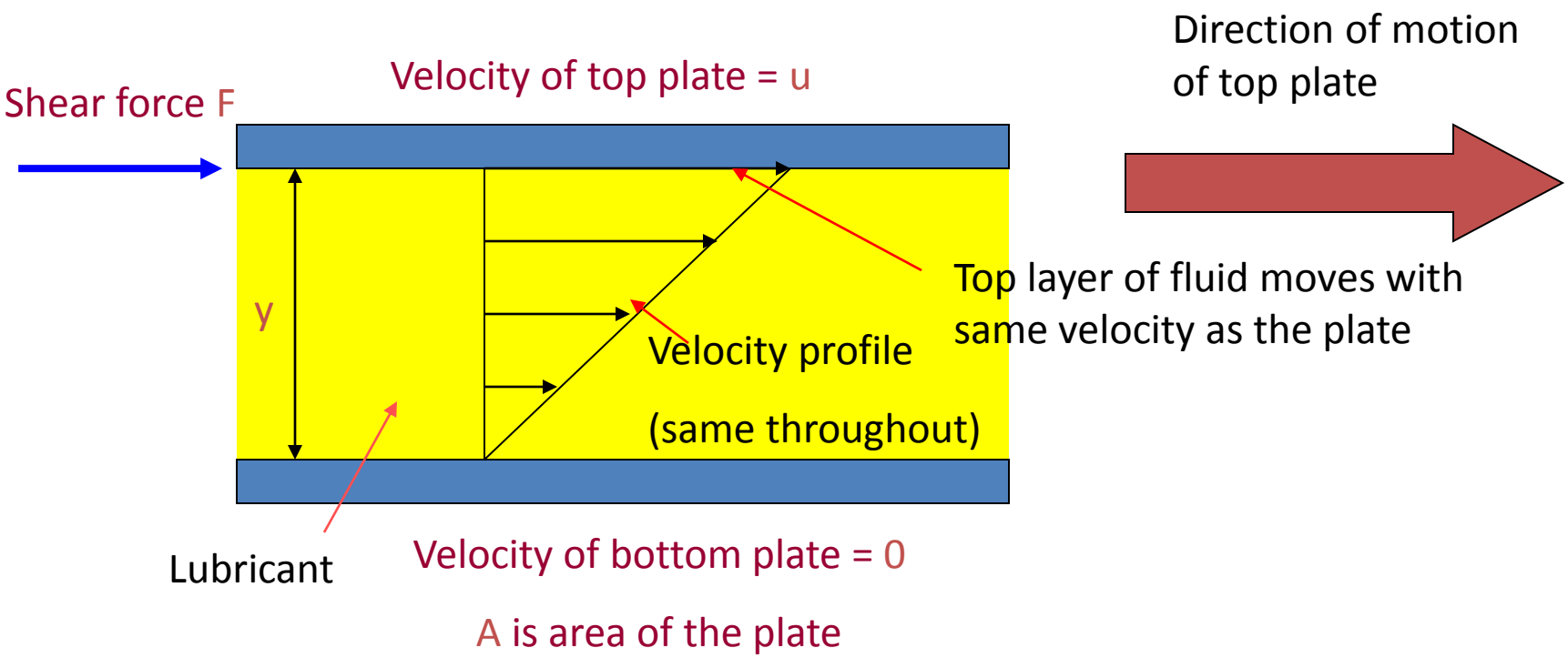






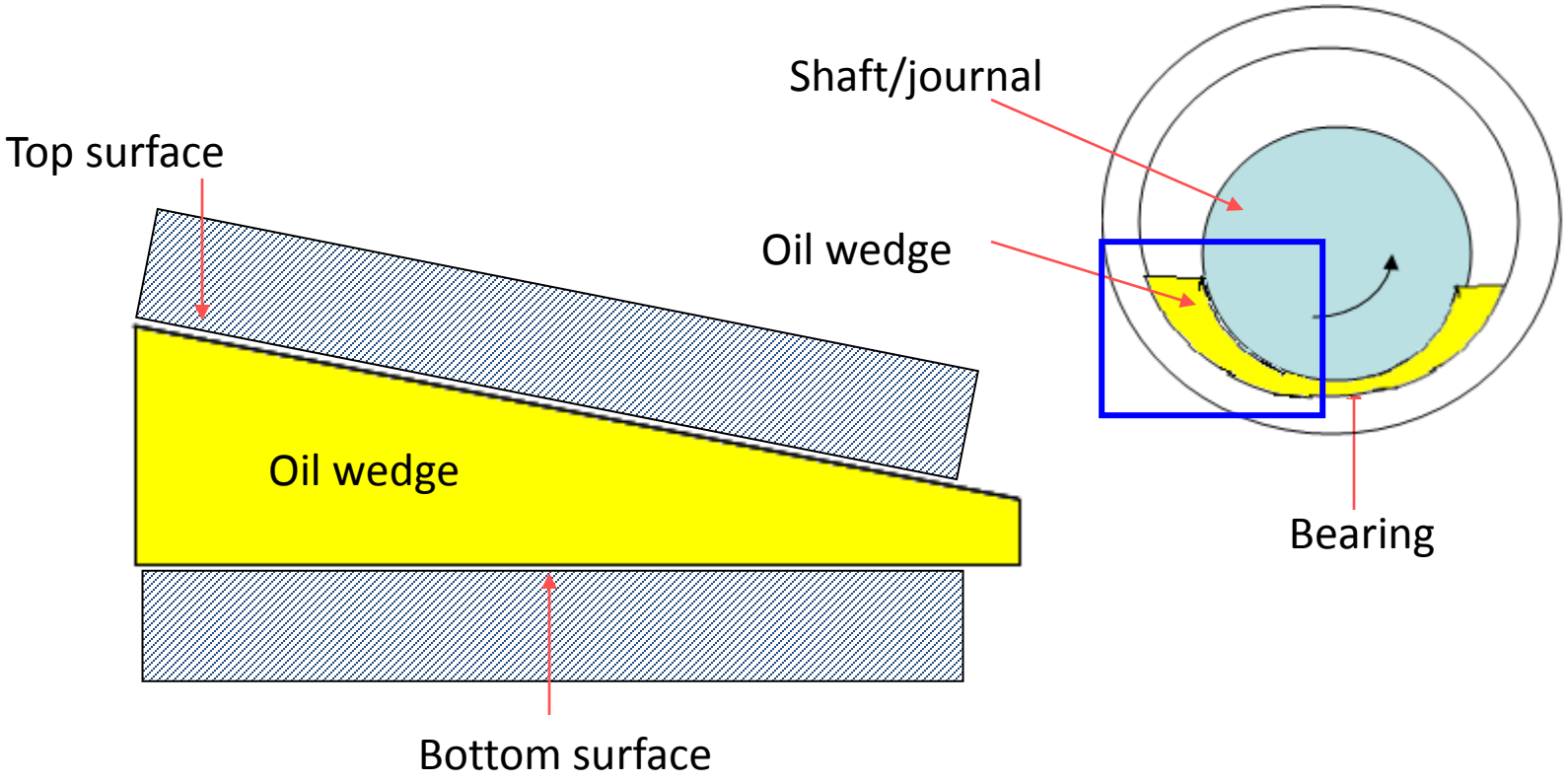


Parallel surfaces



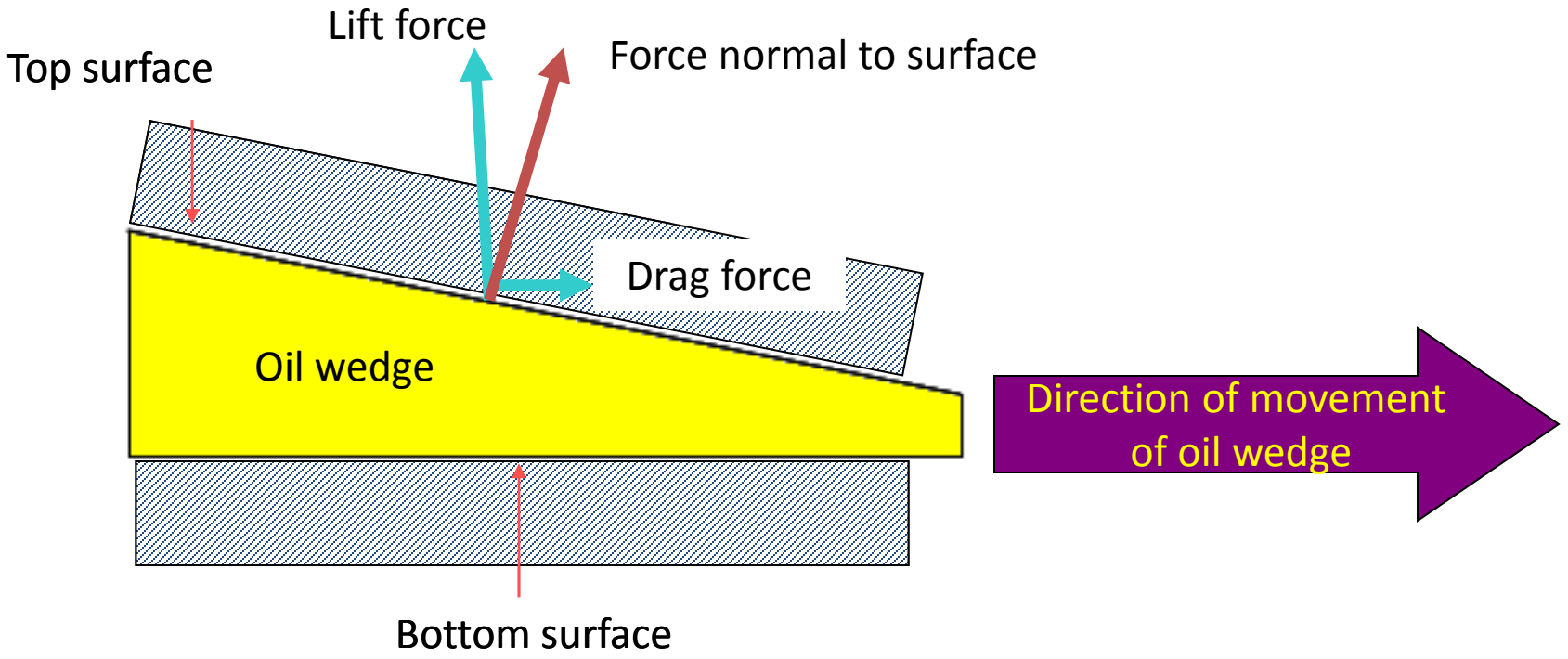
- **There is no pressure buildup in the fluid due to relative motion**
- **It remains constant throughout, influenced only by the load**
- **As load increases the surfaces are pushed towards each other until they are likely to touch.**

Hydrodynamic theory- journal bearings



Oil wedge forms between shaft/journal and bearing due to them not being concentric

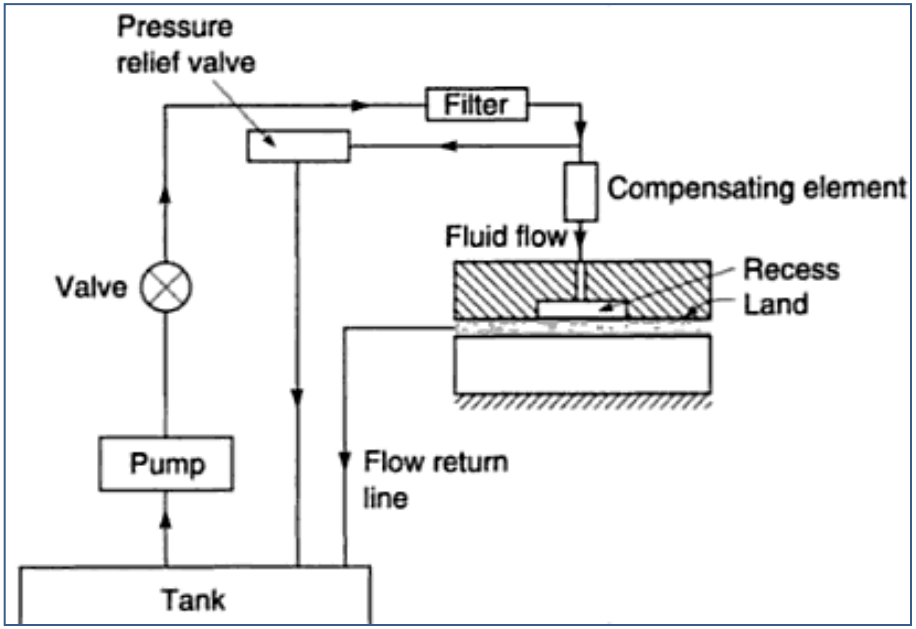
Hydrodynamic lubrication



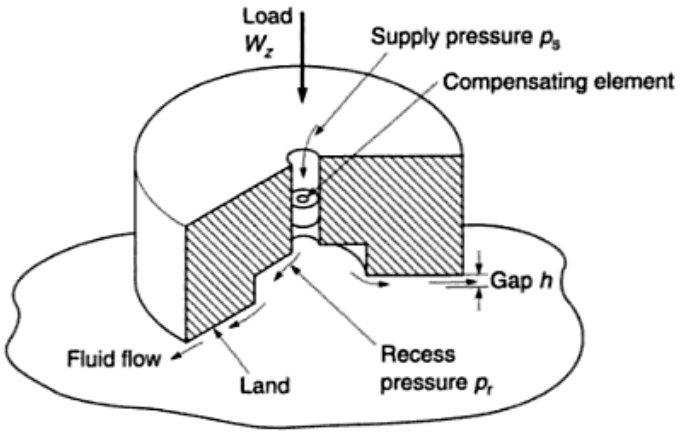
- Surfaces are inclined to each other thereby compressing the fluid as it flows.
- This leads to a pressure buildup that tends to force the surfaces apart
- Larger loads can be carried

2. Hydrostatic Lubrication

- Hydrostatic lubrication is characterized by the complete separation of the conjugated surfaces of a kinematic pair by means of a film of fluid which is pressurized by an external equipment like a pump.
- Its distinguishing features are lack of wear, low friction, high load capacity, a high degree of stiffness and the ability to damp vibrations.
- It differs from hydrodynamic lubrication that the contact pressure is generated by an external pump instead of by viscous drag.



Fluid supply system



Hydro static thrust bearing with circular step pad

General Characteristics of hydrostatic bearings

- A complete lubricant film can be present at even zero sliding speed due to the external pressurization.
- Hydrostatic films usually have thickness upto 100 microns.
- Large starting friction problems are totally eliminated in hydrostatic lubrication.
- Hydrostatic bearings possess zero static friction and high load capacity and stiffness
- The highest axial bearing stiffness, self-aligning capability and the minimal flow is obtained with the smallest film thickness.

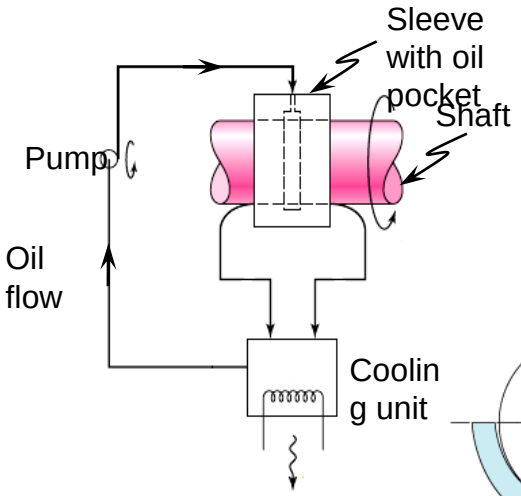
Hydrodynamic bearings assisted by an externally pressurized lubricant supply are known as hybrid bearings

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HYDROSTATIC LUBRICATION

• It is obtained by introducing the lubricant, which is sometimes oil or air or water, into the load-bearing area at a pressure high enough to separate the surfaces with a relatively thick film of lubricant.(Fig. 3a)

• So, unlike hydrodynamic lubrication, this kind of lubrication does not require motion of one surface relative to another.



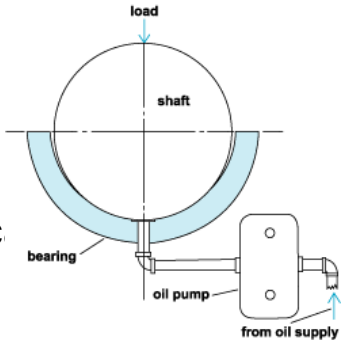


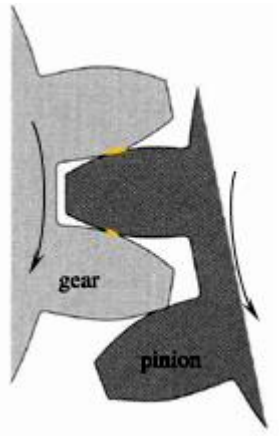
Fig.3a Hydrostatic lubric system

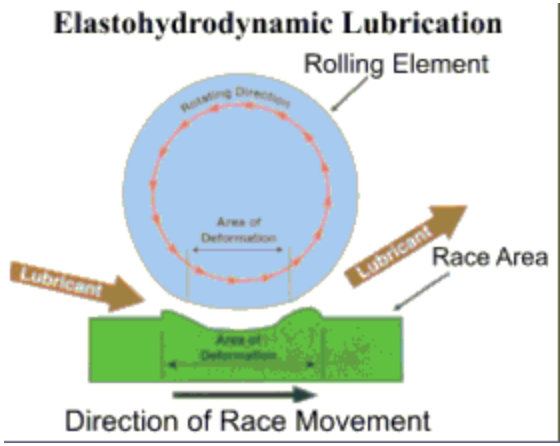
Example: the cross head pin bearing or gudgeon pin bearing in two stroke engines employs this hydrostatic lubrication mechanism. Vertical turbo generators, Centrifuges and ball mills

3. ELASTO-HYDRODYNAMIC LUBRICATION

• It is the phenomenon that occurs when a lubricant is introduced between surfaces that are in rolling contact, such as mating gears (Fig.3b) or rolling bearings.

• The mathematical explanation requires the Hertzian theory of contact stress and fluid mechanics.

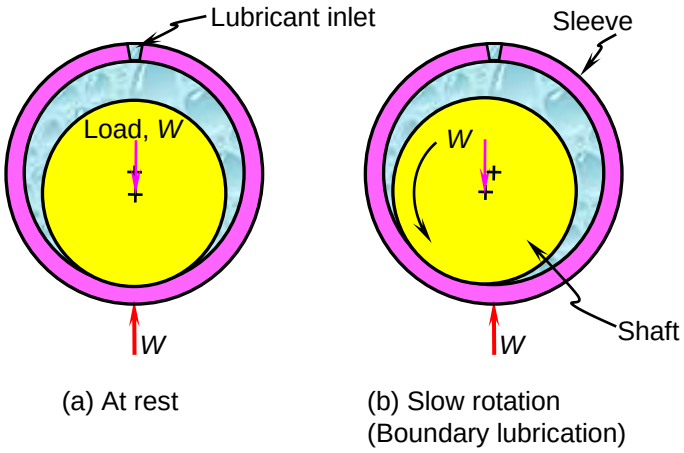


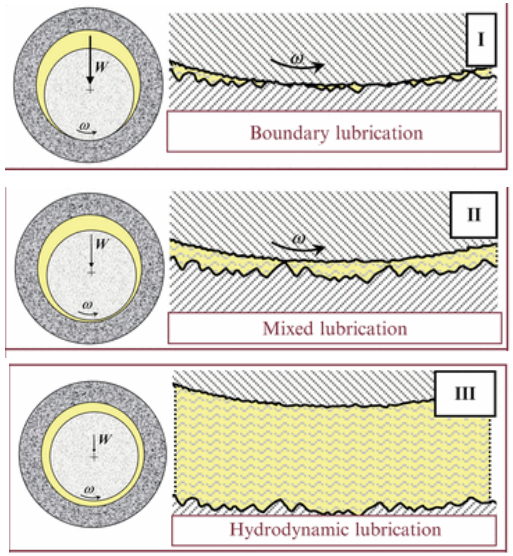


Elasto-dynamic lubrication system

4. BOUNDARY LUBRICATION

- Insufficient surface area, a drop in the velocity of the moving surface, a lessening in the quantity of lubricant delivered to a bearing, an increase in the bearing load, or an increase in lubricant temperature resulting in a decrease in viscosity—any one of these—may prevent the buildup of a film thick enough for full-film lubrication.
- When this happens, the highest asperities may be separated by lubricant films only several molecular dimensions in thickness. This is called boundary lubrication.
- The change from hydrodynamic to boundary lubrication is not at all a sudden or abrupt one.
- It is probable that a mixed hydrodynamic- and boundary-type lubrication occurs first, and as the surfaces move closer together, the boundary-type lubrication becomes predominant.





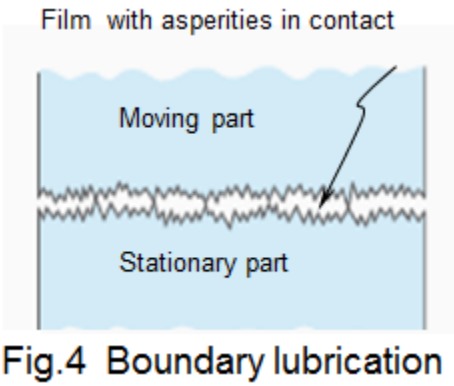
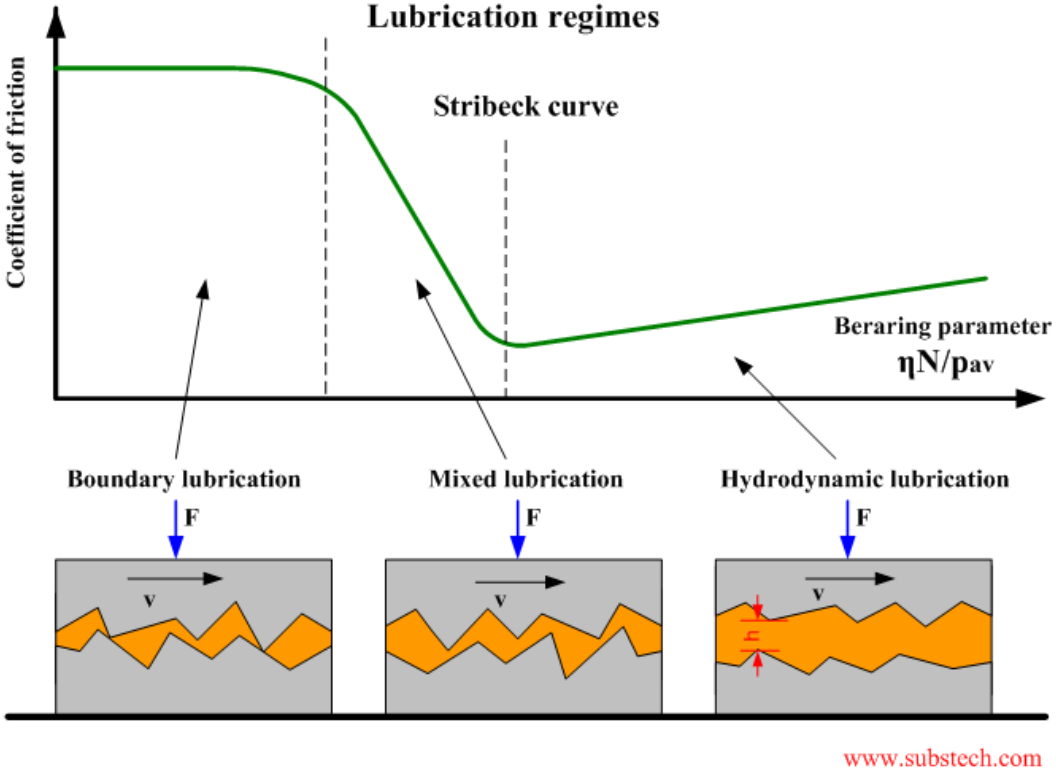
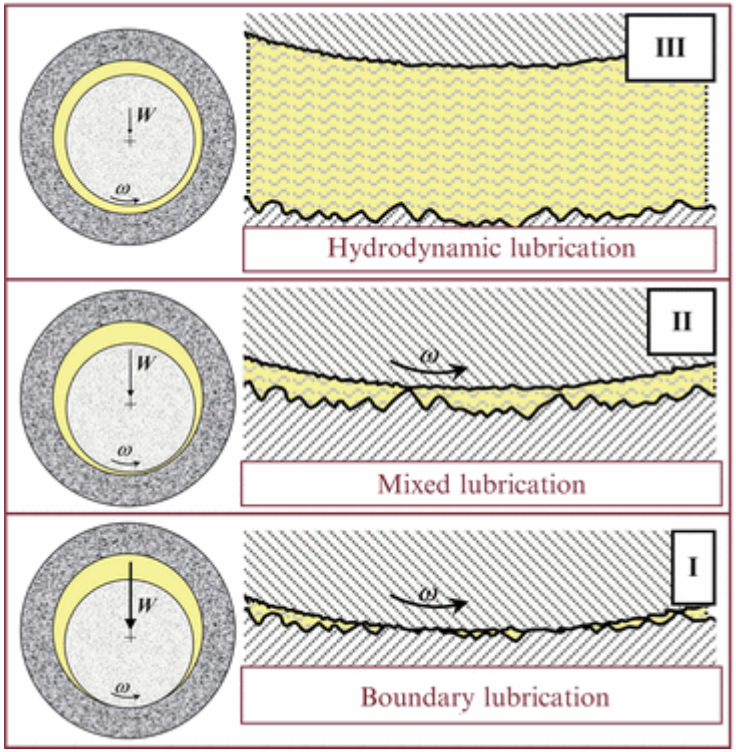


Fig.4 Boundary lubrication

Lubrication Regimes





Examples for boundary lubrication:

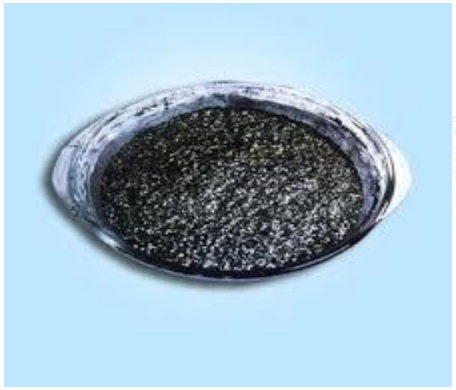
•Guide and guide shoe in two stroke engine. Lubrication of the journal bearing in diesel engines (mainly during starting and stopping of engine).

•Piston rings when cylinder liner is at TDC and BDC position when the piston direction changes and if the relative speed is very slow.

5. SOLID-FILM LUBRICATION

- When bearings must be operated at extreme temperatures, a solid-film lubricant such as *graphite or molybdenum disulfide* must be used because the ordinary mineral oils are not satisfactory.
- Much research is currently being carried out in an effort to find composite bearing materials with low wear rates as well as small frictional coefficients.









Graphite











Molybdenum Disulfide